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| SESSION GUIDE<br><b>CONSTRUCTION SCHEDULING</b><br><i>Topping Out</i> — a week-by-week scheduling game — <i>Racer Commons</i> , 3-story commercial, 38 activities |                                      |                              |                              | SHEET<br><b>IG-01</b>             |
| DEMO<br><b>10 of 17 · Part four</b>                                                                                                                               | COURSE LEVEL<br><b>First-year CM</b> | SESSION<br><b>75 minutes</b> | TEAMS<br><b>3–4 students</b> | PREREQ<br><b>Network diagrams</b> |

Every team schedules the same building, week by week, from the same seed code. The seed fixes the project, the sequence of events *and* the roll behind every judgment call before anyone touches anything, so two teams that finish \$200,000 apart did not get different weather and did not get different luck — they bought different exposure. That is the whole pedagogical trick. You need: one laptop per team (phones work), a projector, and a seed code on the board.

**This sheet is the session.** The CPM derivations, the full calls deck, all eleven debrief questions, the seed table, the overtime literature and the vocabulary live in the **CPM Appendix** — linked from the Guide and from Settings, and not reproduced here. Nothing was dropped in the split; it moved.

## 1 LEARNING GOALS

Not stated objectives — what the activity itself demonstrably shows. One per stage, and these are the four Takeaway cards the learner reads on screen, verbatim:

1. Float is computed, not declared, and the chain with none is the one that sets the finish date.
2. The critical path is re-solved every Monday against what actually happened, so it moves under you.
3. A call is a priced bet with its odds printed on it; every team faces the same roll, so what differs is the exposure you bought.
4. A decision is graded on what it was worth the Monday you made it, not on how the roll landed.

## 2 WHERE THIS SITS ON THE PATH

Demo 10 of 17, Part four — example projects. Part four shows the same habit of mind applied in a real discipline: state what you expect, run the thing, and let the result correct you. This one is the scheduling case, and the prerequisite it leans on is the network diagram — activities, predecessors, durations. It does not need any of the AI demos.

## 3 ONE QUESTION FOR THE SESSION

**What was that decision worth on the Monday you made it?**

*“Nobody in this room is going to be graded on whether the roll went your way. You are going to be graded on what you knew, what it cost, and where the float was when you decided.”*

## 4 SEVENTY-FIVE MINUTE RUNNING ORDER

The tutorial project, *Shoe Tree Annex*, is 10 activities, runs 8–10 turns and carries three calls of its own. **Do not skip it.** Teams that meet the interface on the graded run spend six weeks clicking instead of thinking, and the debrief gets thin.

| MINUTES | LENGTH | SEGMENT                                                                                                                                                                                                                                                                                                         |
|---------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0–5     | 5 min  | <b>Setup</b> (Stage 1). Seed code and difficulty on the board; teams enter both and name themselves. Presentation mode up front. Confirm every team reads the same project start date, Mon 7 Sep 2026, before anyone commits a week.                                                                            |
| 5–11    | 6 min  | <b>CPM at the board.</b> Forward pass, backward pass, total float, on the ten-activity tutorial network — the hand calculation is in the Appendix, §1. Land one sentence: float is not slack in your day, it is what a specific activity can slip before it moves the finish. Leave free float for the debrief. |
| 11–22   | 11 min | <b>Tutorial run</b> — 8–10 turns. Walk the room. Push teams to read the float column before deciding anything, and make sure every team answers the subgrade call instead of clicking past it.                                                                                                                  |
| 22–26   | 4 min  | <b>Brief the competition.</b> Same seed, same difficulty, score is profit, no undo. Say two things out loud: calls default to the free option, so not answering is answering; and crews take two weeks, so manpower is a forecast.                                                                              |
| 26–59   | 33 min | <b>Competitive run</b> (Stages 2–3). Twenty-four to thirty turns. Roughly half the weeks are one click; the rest carry a call, a weather decision or a crew order. Checkpoint at minute 44 — teams should be near week 12, and anyone behind should commit the quiet weeks fast.                                |
| 59–63   | 4 min  | <b>Leaderboard and the do-nothing baseline.</b> Show the spread, then what absorbing everything would have scored on this seed. Worst team against baseline is usually more interesting than first against second.                                                                                              |
| 63–75   | 12 min | <b>Debrief</b> (Stage 4). Open on the calls table; it is the fastest route into the room. Then the five misconceptions below and two or three questions from Appendix §5. Finish on the week the critical path moved and ask who noticed at the time.                                                           |

#### IF YOU ARE SHORT ON TIME

Call time at minute 59 wherever teams are and score *projected* profit — the CPM finish date priced out with general conditions, liquidated damages and the early bonus. It is a legitimate score and it ranks teams the same way. Alternatively, assign the tutorial as ten minutes of homework and give the whole 44 minutes to *Racer Commons*. Do not buy time by cutting the debrief: the calls table is where the session pays off.

## 5 STAGE BY STAGE

### STAGE 1 — READ THE SCHEDULE

**Question.** Where is this job's float, and which chain has none?

**Predict prompt.** Before they start the job: *which chain has zero float?* The prediction is captured on screen and echoed beside the computed answer.

**Activity.** Seed and difficulty in, **Start the job**, then read the **Gantt** and **Network** views before committing anything. 4–5 minutes.

**Expected observation.** Unimpeded, the network runs **89 working days against a 134-day contract** — 45 working days, 63 calendar days, of cushion. One chain carries zero total float and everything else has some.

**Answer key.** The float column, not the bar length, identifies the critical chain. A bar can be long and have float; a two-day cure can be short and have none.

**Teaching limit.** Float here is measured to the project's own computed finish, so it is never negative. Students who have seen negative float in P6 need that said explicitly.

### STAGE 2 — RUN A WEEK

**Question.** What does one week actually change?

**Predict prompt.** *Which activity is critical after this week?* — asked **before** Commit and captured, because the engine resolves the network afterwards and will answer it. This is the demo's signature prediction; make the room commit out loud too.

**Activity.** Read **THE NEWS**, allocate crews, **Commit the week**, watch five working days run in 4D. 2 minutes for the first one, seconds thereafter.

**Expected observation.** The prediction appears beside the computed result. Sooner or later they disagree, and the game announces the move and names the activity it moved to.

**Answer key.** Completed work is pinned to the dates it actually happened, work in progress is anchored to its real start, and nothing not yet started can begin before today. Criticality falls out of that.

**Teaching limit.** One turn is one week of five working days; the engine has no clock, no network call and no unseeded randomness.

### STAGE 3 – MAKE A CALL

**Question.** What are you buying when you answer a call?

**Predict prompt.** Before choosing: *what do you think the free option costs you, in dollars?*

**Activity.** Answer the call on the table — each option shows its price today and its **exposure**, the probability the deferred consequence fires — then **Commit**. 16 calls in *Racer Commons*, 3 in the tutorial.

**Expected observation.** Nothing is hidden: the card prints the exposure as a percentage and names the activity where the consequence will land. Buying exposure down is buying a smaller bar for the same roll.

**Answer key.** Every call defaults to its first option, which is always free and always the most exposed. Clicking Commit through the job takes the 45%, the 40% and all the rest without deciding to.

**Teaching limit.** Exposure is a stated probability against a seeded roll, not a model of real construction risk. One option is deliberately **△ UNSAFE** and is *not* a trade-off — see the box below.

### STAGE 4 – DEBRIEF

**Question.** Was that a good decision, or a good outcome?

**Predict prompt.** Before opening it: *will your calls table read mostly “covered” or mostly “got away with it”?*

**Activity.** Open the **Debrief**. One row per call: Week / Call / You chose / Cost / Exposure / The roll / Outcome. 12 minutes with the room.

**Expected observation.** The verdicts are a two-by-two, and the diagonal is the lesson: *got away with it* is where the luckiest team lives, and *unlucky: you covered it and it still bit* is where a team that did everything right can be sitting.

**Answer key.** Same seed, same roll. Put a team that left a call at 45% beside a team that bought it down to 3% and price both on the Monday they decided.

**Teaching limit.** Profit is the score, but the cash curve is the second set of books — a team can project a healthy profit while sitting \$1,000,000 into the bank's money.

#### THE ONE CALL THAT IS NOT A TRADE-OFF

Around week six, backfilling against green unbraced foundation walls is offered three ways. Waiting costs three days; engineered kickers buy them back for \$6,400; and backfilling high against one side now is free, saves the three days and is marked **△ UNSAFE**, with the reason printed before anybody clicks. An unbraced, uncured wall loaded on one side only is an OSHA excavation and structural-stability violation before it is a schedule decision.

**It is in the deck deliberately**, and if a team takes it the debrief opens with a red block *above* their profit — whether or not the wall stood. If the roll went their way it says so: *and it held. That is the most dangerous possible outcome, because it teaches you it was fine*. Read the block, then the number, then ask whether they would sign that decision with their own name on the citation.

## 6 FIVE MISCONCEPTIONS, AND THE MOMENT THAT EXPOSES EACH

These are the on-screen refutation bank too: each stage's **Check yourself** card offers the flawed model as a real option and names why it is wrong. State the misconception before the correction — a correction alone does not displace a model the student already holds.

**(a) “Float is free time you can spend casually.” — Stage 1**

Total float belongs to the whole chain and is consumed by whoever spends it first.

**MEP1**

carries 9 days of total float but only

**1 day of free float**

: slip it three days and the finish has not moved, but Level 2 rough-in cannot start when it wanted to. Then the switchgear card takes two days and the failed rough-in inspection takes three, on a branch that had four.

**(b) “Overtime always accelerates.” — Stage 2**

It genuinely does at first — a 50-hour week delivers 1.25× and a 60-hour week 1.50× in week one, and that is real acceleration, not a concession. But the cost premium applies in full from week one and never decays, while efficiency does: by week 5 a 50-hour crew produces

**0.938**

of a straight-time week at

**1.375×**

the labour cost. Overtime is a burst tool for a defined push. And it does nothing at all to a cure or a hold.

**(c) “The critical path is fixed at baseline.” — Stage 2**

It is recomputed every Monday.

**ROOF**

sits off the path at baseline with six days of float; two rain days plus the membrane-backorder card and it is critical — at which point the glazing card that arrives later matters twice as much. The debrief question is not whether they saw the announcement, it is whether anyone changed their allocation the week *before*.

**(d) “Weather delays everything equally.” — Stage 3**

Rain days stop only weather-flagged work: earthwork, concrete placement, roofing and envelope. The same “ground is soup” card costs three critical days in week 3 and

**exactly nothing**

in week 18 with the building dried in. A cure keeps curing through a washout. The forecast is not the decision — the float column is.

**(e) “A good outcome means it was a good call.” — Stage 4**

The gambler's fallacy in a hard hat, and the most expensive habit on this list. A team pours to plan at column line 4, the roll comes in at 0.61, the inspector takes the pads, and they leave convinced they read the geotech. They took a 45% chance of a three-day hold and \$8,500 — and the roll every other team was handed landed above the line.

## **7 PRESENTER CONTROLS**

**Settings** (⊗ in the header) holds **Open Presenter Notes**, **Presentation mode** and **Reset**, in that order, then this demo's own controls: the Guide, the printable guide and the **CPM Appendix**, the engine self-test, and clearing the leaderboard.

**Presentation mode** — projector mode — scales the page up and puts a Notes button in the header. Turn it on for THE NEWS each week so the whole room reads the same card, and for the leaderboard and calls tables at minute 59. It survives Reset deliberately; a presenter resets between rooms.

**Reset** puts the job back to the setup sheet in place, with no page reload: no run in progress, seed and team cleared, every prediction echo and check answer cleared, stage 1 selected. Presentation mode and the leaderboard are not

demo state and survive it. Mid-run it asks first, because a reset throws away a class's work.

**Engine self-test** runs **36 assertions** live — the tutorial CPM against the hand calculation, determinism, the non-compressible holds, the weather boundary in both directions, and the two checks that hold the calls fair. If a student says the software is wrong, run it in front of them; it is a surprisingly good two minutes of teaching.

**If a score is disputed or a browser dies.** Same seed plus same decisions produces the same profit to the dollar, and that is one of the 36. Replay the seed with the team's decision log.